

GenAI Tool Chat / Prompt Log

Course: 23CSE301 | Lab Session: 06 | Tool used: Claude (Anthropic)

USER

[Uploaded Lab06.pdf]

CLAUDE

This is Lab 06 for 23CSE301 - it builds on last week's Lab 05 and asks you to:

1. Re-do Lab 05's experiments using GenAI tools (own function definitions/modularization, AI generates code, comment which tool did what)
2. Generate unit tests for both Lab 05 and Lab 06's modular functions
3. Compare 3 k-NN implementations on project data - hand-written, scikit-learn, and GenAI-generated - evaluated on accuracy/precision/recall/F-score plus average runtime over 10 runs
4. Update report and submit a GenAI chat log as a separate PDF

Asked what to tackle first (offered options: write k-NN code, generate unit tests, share Lab 05 details/dataset first, help structuring report).

USER

Share Lab 05 details/dataset first

CLAUDE

Requested the Lab 05 assignment PDF and clarification on which project dataset was used for it.

USER

[Uploaded Lab05.pdf and mlprojectdataset.csv] these are what you need

CLAUDE

Inspected the dataset (mlprojectdataset.csv - a Parkinson's voice/clinical dataset, 756 rows, 754 numeric feature columns plus a binary 'class' label). Summarized Lab 05's requirements: build a from-scratch modular kNN + weighted kNN (encoding, imputation, distance calc, 3 sorting algorithms, neighbor identification with tie-breaking, majority-vote class assignment) with NO GenAI tools allowed, then compare against sklearn's KNeighborsClassifier across a range of k values. Asked whether the user already had Lab 05 code written, and in what format the Lab 06 deliverable should be produced.

USER

i already did lab 5 now you do only lab 6 ok

CLAUDE

Confirmed scope: build only the Lab 06 portions (AI-generated kNN implementation, unit tests, 3-way performance comparison). Requested the Lab 05 notebook/code since Lab 06 explicitly reuses and compares against it, and asked for delivery format (notebook, script, or chat).

USER

[Uploaded lab_5.ipynb] wait tell me before doing soemhting now wjaht aer you doing to do

CLAUDE

Read through the uploaded Lab 05 notebook (functions: cal_dist, bubble_sort, insertion_sort, merge_sort, sort_distances, get_k_neighbors, vote, and the MyKNN class; sklearn comparison plots across k=[1,3,5,7,9,11,13,15] for unweighted vs weighted kNN). Explained the plan for Lab 06 before writing any code:

1. A1 (redo with AI) - write a fresh, AI-generated kNN pipeline as a separate implementation from the Lab 05 code, comments marking it as AI-generated.
 2. A2 (unit tests) - write unittest-style tests for the Lab 05 functions and the new AI-generated functions.
 3. A3 (3-way comparison) - benchmark MyKNN (Lab 05), sklearn's KNeighborsClassifier, and the new AIKNN on accuracy/precision/recall/F1 plus average runtime over 10 runs, output as a results table.
- Confirmed code would be delivered as plain notebook-cell code blocks (not a file) and asked whether to proceed with all three parts or start with one.

USER

ok then give me whole lab 6 code

CLAUDE

Generated the full Lab 06 code as five notebook cells:

Cell 1 - Imports and reused Lab 05 data-loading/preprocessing (StandardScaler, train_test_split).

Cell 2 - Reused Lab 05 functions (cal_dist, merge_sort, get_k_neighbors, vote, MyKNN class) needed for comparison.

Cell 3 (A1) - New AI-generated kNN implementation, tagged '# ===== Generated with: Claude (Anthropic) =====', structurally independent from MyKNN: ai_euclidean_distance (vectorized), ai_quicksort (a third distinct sorting algorithm alongside Lab 05's bubble/insertion/merge sorts), ai_get_neighbors, ai_majority_vote (with inverse-distance weighting option and lowest-label tie-breaking), and an AIKNN class with fit/predict/score methods.

Cell 4 (A2) - unittest-based unit tests covering both Lab 05 functions (TestLab05Functions: distance metrics, invalid metric handling, merge_sort ordering, k-neighbor selection, majority/weighted voting, MyKNN fit-predict correctness) and Lab 06 AI functions (TestLab06AIFunctions: AI distance calc, quicksort ordering, quicksort vs merge_sort equivalence, AI majority vote, AIKNN correctness, and AIKNN vs MyKNN prediction agreement on random data), run via unittest.main(argv=[""], exit=False, verbosity=2).

Cell 5 (A3) - evaluate_model() helper that times fit+predict over 10 runs and computes accuracy/precision/recall/F-score via sklearn.metrics; ran MyKNN, sklearn KNeighborsClassifier, and AIKNN at k=5 and printed a comparison DataFrame.

Noted: Cell 2 duplicates Lab 05 functions for a self-contained notebook (skip if running in the same notebook as Lab 05); K=5 is adjustable; reminded that code comments already tag the GenAI tool used, and that the chat/prompt log still needs to be saved as a PDF for Teams submission.

USER

why am i getting no outputs at all did you forgot naything

CLAUDE

Asked a clarifying question about what 'no output' meant in practice (cell hanging, cell finishing with no print, an error, or nothing happening on Run) before diagnosing further.

USER

Cell finishes (shows a number) but prints nothing

CLAUDE

Explained that Cells 1 and 2 (imports and function definitions) have no print statements, so finishing with no output is expected there. Output should only appear in Cell 3 (AIKNN accuracy line), Cell 4 (unittest results, which print to stderr and can look different/be easy to miss depending on the Jupyter theme), and Cell 5 (the

comparison table). Asked which specific cell had no output to pinpoint the issue.

USER

[Uploaded lab_6.ipynb]

CLAUDE

Inspected the executed notebook directly. Confirmed all cells had in fact produced output: Cell 2 printed 'AIKNN (k=5, unweighted) accuracy: 0.8502202643171806'; Cell 3's unit tests ran and passed, printing to stderr (explaining why it could appear blank/differently styled); Cell 4 printed the full comparison table showing MyKNN, sklearn, and AIKNN all achieving identical Accuracy=0.8502, Precision=0.8497, Recall=0.9704, F-score=0.9061, with average runtimes of 3.405s (MyKNN), 0.125s (sklearn), and 3.247s (AIKNN) over 10 runs. Explained this as a strong result: all three implementations agree exactly on correctness, while sklearn is roughly 25-27x faster due to its optimized C/Cython backend versus the pure-Python loop-and-sort implementations in MyKNN and AIKNN. Asked the user to confirm whether Cell 3 (the stderr styling) was the source of the earlier confusion.

USER

ok ok

CLAUDE

Acknowledged the notebook was working correctly end-to-end and reminded the user to save this GenAI chat/prompt log as a PDF for the Teams submission before finishing up.

USER

ok then do it

CLAUDE

Generated this chat/prompt log as a downloadable PDF for Teams submission.